

BLUESTOCK FINTECH
MUTUAL FUND ANALYTICS CAPSTONE PROJECT
Final Technical Report

Submitted by
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Acknowledgement

I would like to express my sincere gratitude to Bluestock Fintech for providing me with the opportunity to work on the Mutual Fund Analytics Capstone Project. This project allowed me to apply concepts of data engineering, data analysis, financial analytics, and business intelligence using real-world mutual fund datasets.

I am thankful to my mentors and the Bluestock team for their guidance, project documentation, and continuous support throughout the internship. The project helped me strengthen my practical knowledge of Python, SQL, Power BI, data visualization, and financial performance analysis.

Finally, I would like to thank everyone who contributed directly or indirectly to the successful completion of this project.

Abstract

This project presents an end-to-end Mutual Fund Analytics solution developed using Python, SQLite, SQL, Jupyter Notebook, and Power BI. The objective was to build a complete analytics pipeline capable of ingesting, cleaning, storing, analyzing, and visualizing mutual fund data to generate actionable insights.

The project includes ETL processes, database design, exploratory data analysis, performance analytics, advanced risk analytics, investor behavior analysis, and an interactive dashboard. Financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), and Sector Concentration (HHI) were implemented. Additional analyses included investor cohort analysis, SIP continuity analysis, and a mutual fund recommendation engine based on risk appetite.

The final outcome is a comprehensive analytics solution that demonstrates practical skills in data engineering, financial analysis, and business intelligence while providing meaningful insights into mutual fund performance and investor trends.

1. Introduction

The Mutual Fund Analytics Capstone Project is an end-to-end data analytics solution developed to analyze various aspects of the Indian mutual fund industry. The project integrates data engineering, database management, financial analytics, and business intelligence techniques to transform raw mutual fund datasets into meaningful insights.

The project utilizes multiple datasets, including fund master information, historical NAV data, investor transactions, SIP inflows, AUM statistics, benchmark indices, portfolio holdings, and scheme performance. These datasets were cleaned, validated, and stored in a structured SQLite database before performing exploratory data analysis, financial performance evaluation, and advanced analytics.

The solution was developed using Python for data processing, SQLite for database management, SQL for querying, Jupyter Notebook for analysis, and Power BI for interactive dashboard development. The final outcome provides valuable insights into mutual fund performance, investor behavior, portfolio diversification, and risk analysis while demonstrating practical skills in data engineering and financial analytics.

2. Problem Statement

The Indian mutual fund industry generates a large volume of financial and investor-related data from multiple sources. However, raw datasets often contain inconsistencies, missing values, duplicate records, and different formats, making meaningful analysis difficult.

The objective of this project is to build a complete analytics pipeline capable of:

- Collecting and validating mutual fund datasets.
- Cleaning and transforming raw data into analysis-ready datasets.
- Storing cleaned data in a relational database.
- Performing financial performance analysis using industry-standard metrics.
- Identifying investor trends and behavioral patterns.
- Building interactive dashboards for business decision-making.
- Developing advanced analytics models for risk measurement and fund recommendation.

The project aims to provide a comprehensive analytics solution that supports data-driven investment analysis and business intelligence.

3. Objectives

The primary objectives of this project are:

- Design an end-to-end ETL pipeline for mutual fund datasets.
- Clean and validate multiple financial datasets using Python.
- Create a structured SQLite database with normalized tables.
- Perform exploratory data analysis to identify investment trends.
- Calculate financial performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- Perform advanced risk analytics including VaR, CVaR, Rolling Sharpe Ratio, and Sector HHI Concentration.
- Analyze investor behavior using cohort analysis and SIP continuity analysis.
- Develop a mutual fund recommendation engine based on investor risk appetite.
- Design an interactive Power BI dashboard to visualize insights.
- Present findings through technical documentation and business reports.

4. Technology Stack

Technology	Purpose
Python	Data processing and analytics
Pandas	Data manipulation
NumPy	Numerical computations
Matplotlib	Static visualizations
Plotly	Interactive visualizations
SQLite	Database management
SQL	Data querying
Jupyter Notebook	Analysis and experimentation
Power BI	Interactive dashboards
Git & GitHub	Version control
Requests	Live API data fetching

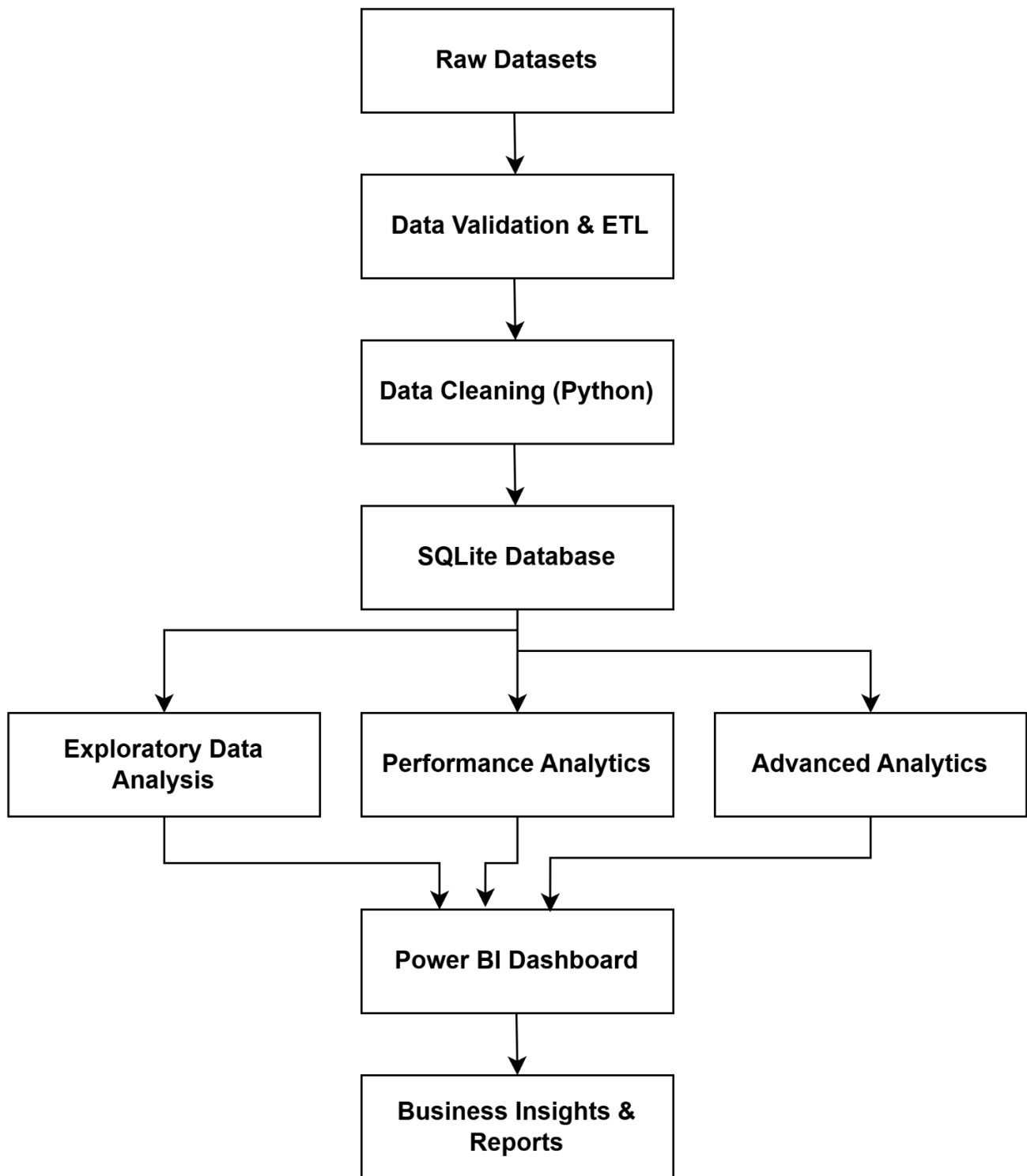


Figure 1: Overall Project Workflow (ETL → Database → Analytics → Dashboard)

5. Dataset Description

The Mutual Fund Analytics project uses multiple datasets representing different aspects of the Indian mutual fund ecosystem. These datasets include scheme information, historical NAV values, investor transactions, SIP inflows, AUM statistics, benchmark indices, portfolio holdings, and scheme performance. Together, they provide a comprehensive view of mutual fund operations, investor behavior, and market performance.

The datasets were sourced from the project resources and were cleaned, validated, and transformed before being used for analysis. Each dataset serves a specific purpose in the analytics pipeline, enabling performance evaluation, risk analysis, investor segmentation, and dashboard visualization.

The following table summarizes the datasets used in the project.

Table 5.1 – Dataset Summary

Dataset	Description
01_fund_master.csv	Mutual fund scheme details including fund house, category, benchmark, expense ratio, and fund manager
02_nav_history.csv	Historical Net Asset Value (NAV) of mutual fund schemes
03_aum_history.csv	Assets Under Management (AUM) data over time
04_monthly_sip_inflows.csv	Monthly SIP inflow statistics and SIP account information
05_category_inflows.csv	Net inflows and outflows across mutual fund categories
06_benchmark_indices.csv	Benchmark index values used for comparison
07_scheme_performance.csv	Fund return and performance metrics

08_investor_transactions.csv	Investor transaction history including SIP, Lumpsum, and Redemption
09_portfolio_holdings.csv	Portfolio holdings of mutual fund schemes
10_industry_folio.csv	Industry folio and investor statistics

Dataset Characteristics

The datasets contain information related to:

- Mutual fund schemes
- Historical NAV values
- Portfolio composition
- Investor demographics
- Transaction history
- Monthly SIP statistics
- Benchmark performance
- Risk and return measures

The data was cleaned and standardized before analysis to ensure consistency, reliability, and accuracy across all analytical modules.

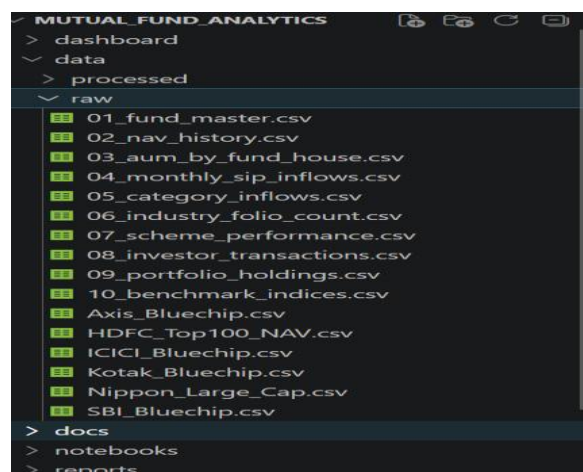


Figure 2: Raw Mutual Fund Datasets Used in the Project

6. Project Architecture

The project follows a modular architecture where data flows through multiple stages, beginning with raw data ingestion and ending with interactive business intelligence dashboards. Each module is designed to perform a specific task, ensuring scalability, maintainability, and reusability.

The workflow begins with raw CSV datasets that undergo validation and cleaning using Python scripts. The cleaned datasets are then loaded into a SQLite database, which serves as the centralized repository for analysis. Exploratory Data Analysis (EDA) is performed using Jupyter Notebook to identify patterns and trends. Financial performance metrics and advanced analytics are then computed before presenting the results through an interactive Power BI dashboard.

The overall workflow is illustrated below.

Project Workflow

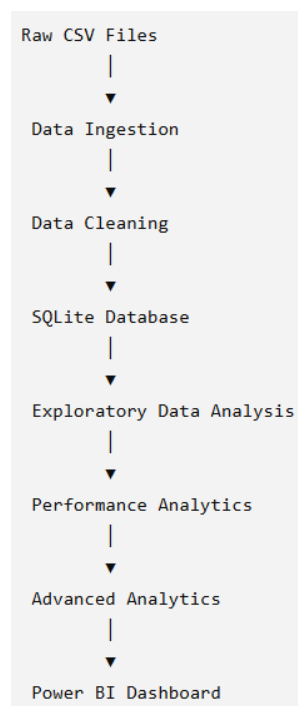


Figure 3: Overall Project Architecture

7. ETL Pipeline

An automated ETL (Extract, Transform, Load) pipeline was developed to process the mutual fund datasets. The pipeline consists of three major stages:

Extract

Raw CSV files containing mutual fund information were collected and validated. Each dataset was inspected for missing values, duplicate records, and inconsistent formats.

Transform

Data cleaning scripts were developed to standardize column names, handle missing values, remove duplicates, convert data types, and validate business rules. Separate Python scripts were used for cleaning each dataset to improve modularity and maintainability.

Load

The cleaned datasets were loaded into a SQLite database with a structured schema. SQL queries were then used for analytical operations and report generation.

The ETL pipeline ensures that all downstream analytics are based on clean, validated, and consistent data.

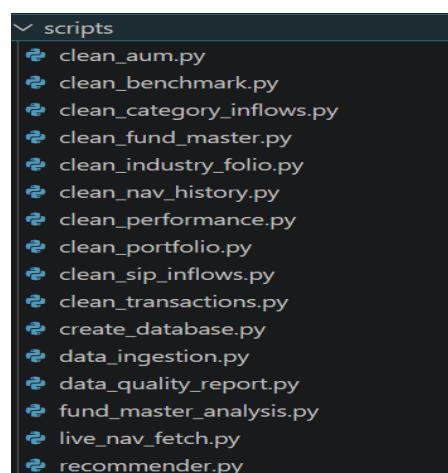


Figure 4: ETL Pipeline Scripts

8. Database Design

A relational database was designed using **SQLite** to store the cleaned mutual fund datasets. The database acts as the central repository for all analytical operations, ensuring efficient storage, retrieval, and querying of data.

Each dataset was imported into a separate table with appropriate data types and primary identifiers. The database design minimizes redundancy while allowing seamless integration of data across different analytical modules.

The database supports SQL-based analysis for performance evaluation, investor behavior analysis, and dashboard reporting.

Table 5.1 – Dataset Summary

Table Name	Description
fund_master	Contains mutual fund scheme details such as scheme name, category, AMC, benchmark, expense ratio, and fund manager.
nav_history	Stores historical Net Asset Value (NAV) for each mutual fund scheme.
aum_history	Contains historical Assets Under Management (AUM) information.
sip_inflows	Monthly SIP collections and active SIP accounts.
category_inflows	Net inflow and outflow data across mutual fund categories.
benchmark_indices	Benchmark index values used for comparison with scheme performance.
scheme_performance	Annualized returns and performance metrics of schemes.
investor_transactions	Investor transactions including SIP, Lumpsum, and Redemption.

portfolio holdings	Equity portfolio holdings with sector allocation and weightage.
industry folio	Industry-level folio statistics and investor counts.

Database Features

The SQLite database provides the following advantages:

- Centralized storage of cleaned datasets.
- Faster execution of analytical SQL queries.
- Improved data consistency.
- Simplified integration with Python notebooks.
- Efficient data retrieval for dashboard development.

```

sql > schema.sql
1  CREATE TABLE fund_master (
2      amfi_code INTEGER PRIMARY KEY,
3      fund_house TEXT,
4      scheme_name TEXT,
5      category TEXT,
6      sub_category TEXT,
7      plan TEXT,
8      launch_date DATE,
9      benchmark TEXT,
10     expense_ratio_pct REAL,
11     exit_load_pct REAL,
12     min_sip_amount REAL,
13     min_lumpsum_amount REAL,
14     fund_manager TEXT,
15     risk_category TEXT,
16     sebi_category_code TEXT
17 );
18
19 CREATE TABLE nav_history (
20     amfi_code INTEGER,
21     date DATE,
22     nav REAL
23 );
24
25 CREATE TABLE scheme_performance (
26     amfi code INTEGER,

```

Figure 5: SQLite Database Schema

9. Exploratory Data Analysis

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the mutual fund datasets and identify meaningful patterns before performing advanced analytics.

Multiple visualizations were created using **Matplotlib** and **Plotly** to analyze investment trends, fund performance, investor demographics, and portfolio composition.

The EDA helped identify important business insights and served as the foundation for performance analytics and dashboard development.

EDA Performed

The following analyses were completed:

- Historical NAV Trend Analysis
- Assets Under Management (AUM) Growth
- SIP Inflow Trend Analysis
- Category-wise Net Inflows
- Benchmark Comparison
- Investor Age Group Distribution
- Gender-wise Investor Distribution
- State-wise Investment Distribution
- Portfolio Sector Allocation
- Correlation Analysis

Key Observations

- Equity-oriented schemes showed higher long-term growth compared to debt-oriented schemes.
- SIP investments demonstrated a consistent upward trend throughout the observed period.
- A significant proportion of investments originated from metropolitan cities.
- Portfolio allocations indicated strong exposure to Banking, IT, and Financial Services sectors.
- Certain mutual fund categories attracted substantially higher inflows than others, indicating investor preference.

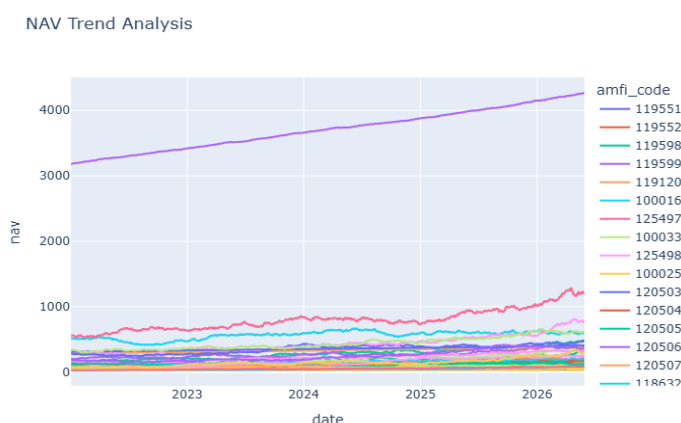


Figure 6: Historical NAV Trend

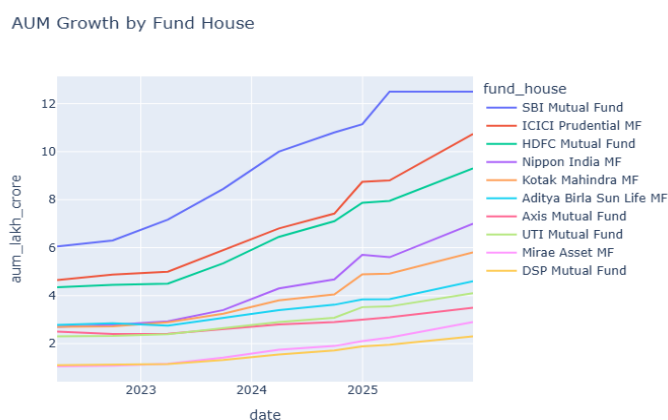


Figure 7: AUM Growth Analysis

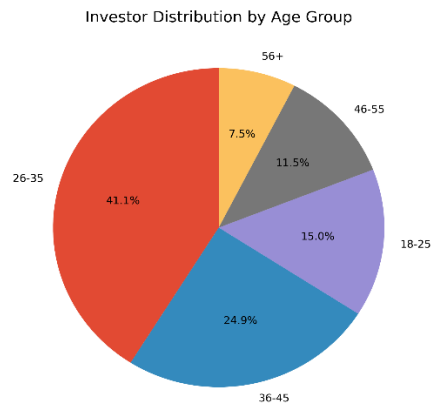


Figure 8: Investor Age Group Distribution

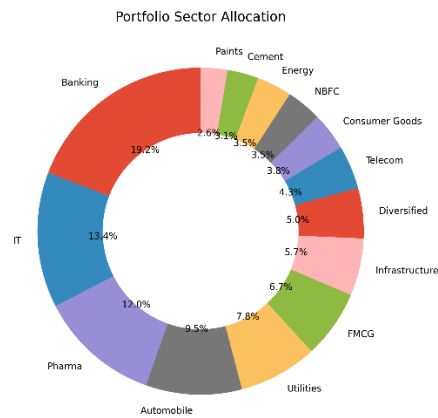


Figure 9: Sector-wise Portfolio Allocation

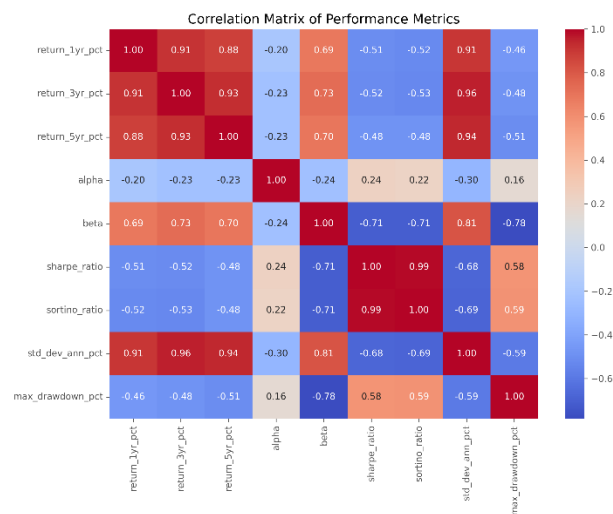


Figure 10: Correlation Matrix of Mutual Fund Metrics

10. Performance Analytics

Performance Analytics

Performance analytics was performed to evaluate the risk-adjusted returns of mutual fund schemes using industry-standard financial metrics. Historical NAV data was used to calculate daily returns, annualized returns, volatility, and several risk indicators. These metrics help investors compare fund performance beyond absolute returns by considering the level of risk associated with each investment.

The analysis also included benchmark comparison and the creation of a consolidated fund scorecard for easier evaluation of mutual fund schemes.

Performance Metrics

The following financial metrics were calculated during the analysis:

- Daily Returns
- Annualized Return (CAGR)
- Annualized Volatility
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

These metrics provide a comprehensive assessment of return generation, downside risk, market sensitivity, and overall investment performance.

Financial Metrics

Daily Returns

Daily return measures the percentage change in a fund's Net Asset Value (NAV) from one trading day to the next. It serves as the foundation for calculating other performance metrics.

CAGR (Compound Annual Growth Rate)

CAGR represents the annualized growth rate of an investment over a specified period, assuming profits are reinvested. It provides a standardized measure for comparing long-term fund performance.

Sharpe Ratio

The Sharpe Ratio measures risk-adjusted returns by comparing excess returns to overall portfolio volatility. Higher values indicate better compensation for the level of risk taken.

Sortino Ratio

Unlike the Sharpe Ratio, the Sortino Ratio considers only downside volatility, making it a more focused measure of downside risk.

Alpha

Alpha measures the excess return generated by a mutual fund relative to its benchmark after adjusting for market risk.

Beta

Beta measures the sensitivity of a fund's returns relative to market movements.

- Beta > 1 indicates higher volatility than the market.
- Beta < 1 indicates lower volatility than the market.

Maximum Drawdown

Maximum Drawdown measures the largest peak-to-trough decline in the value of an investment during the observation period. It helps investors understand the worst historical loss experienced by a fund.

Performance Results

The calculated performance metrics were consolidated into a fund performance scorecard that enables comparison across different mutual fund schemes.

The scorecard combines return-based metrics and risk measures, providing a holistic evaluation of each scheme's historical performance.

Benchmark Comparison

Mutual fund returns were compared against benchmark indices to evaluate their relative performance. This comparison helps determine whether a fund consistently outperformed or underperformed the market during the analysis period.

Key Insights

- Several schemes demonstrated strong long-term returns while maintaining relatively low volatility.
- Higher Sharpe Ratio values indicated better risk-adjusted performance.
- Sortino Ratio analysis highlighted funds with superior downside protection.
- Maximum Drawdown analysis identified schemes that recovered more effectively after market corrections.
- Benchmark comparison revealed that selected schemes consistently outperformed their respective benchmark indices.

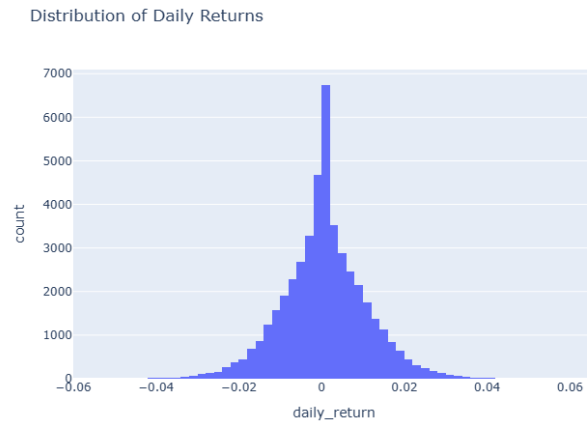


Figure 11: Distribution of Daily Returns

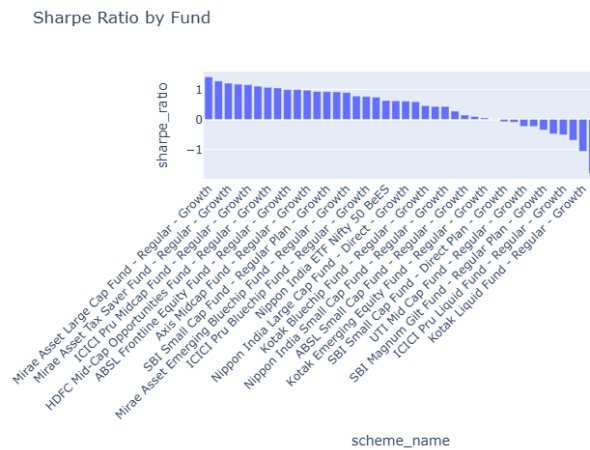


Figure 12: Sharpe Ratio Comparison Across Mutual Funds

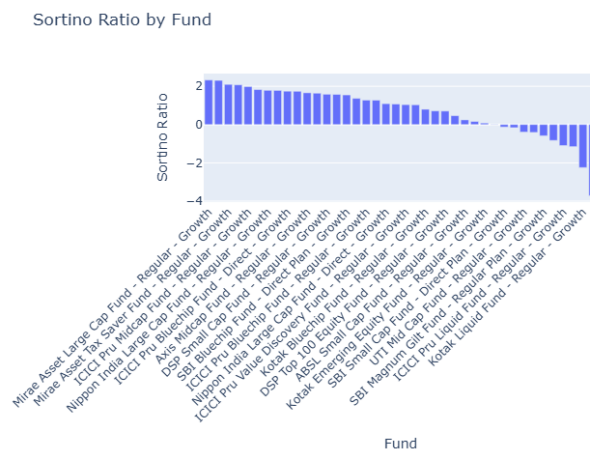


Figure 13: Sortino Ratio Comparison

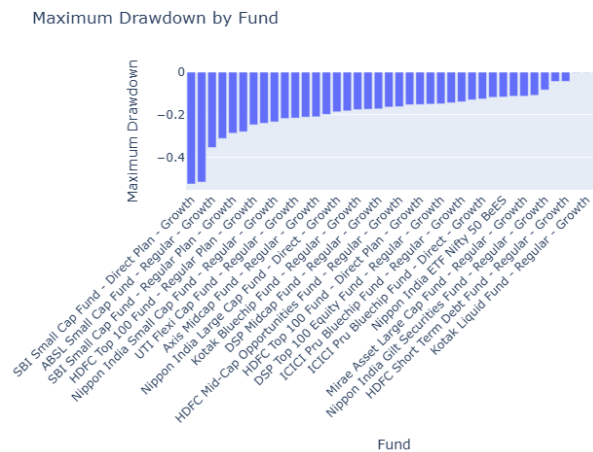


Figure 14: Maximum Drawdown Analysis

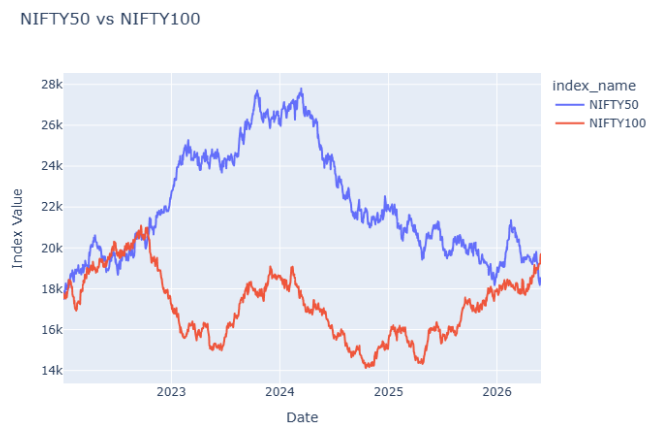


Figure 15: Mutual Fund Benchmark Comparison

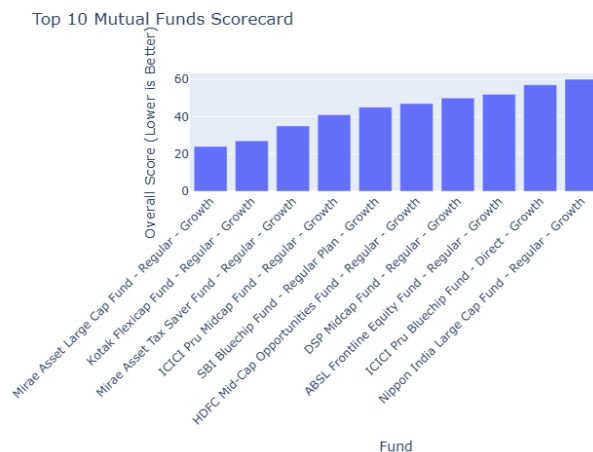


Figure 16: Fund Performance Scorecard

11. Advanced Analytics

Advanced Analytics

The Advanced Analytics module extends traditional mutual fund analysis by incorporating risk measurement, investor behavior analysis, portfolio concentration analysis, and recommendation techniques.

Unlike conventional performance metrics, these analyses provide deeper insights into investment risk, portfolio diversification, and investor engagement patterns.

The advanced analytics module includes:

- Rolling Sharpe Ratio
- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Sector Concentration Analysis (HHI)
- Cohort Analysis
- Investor Churn (At-Risk Investors)
- Mutual Fund Recommendation System

Rolling Sharpe Ratio

Rolling Sharpe Ratio was calculated using a rolling window over historical daily returns to evaluate how the risk-adjusted performance of mutual fund schemes changed over time.

Unlike a single Sharpe Ratio value, the rolling version captures fluctuations in performance under different market conditions.

The analysis enables investors to identify periods of consistent outperformance as well as intervals of increased investment risk.

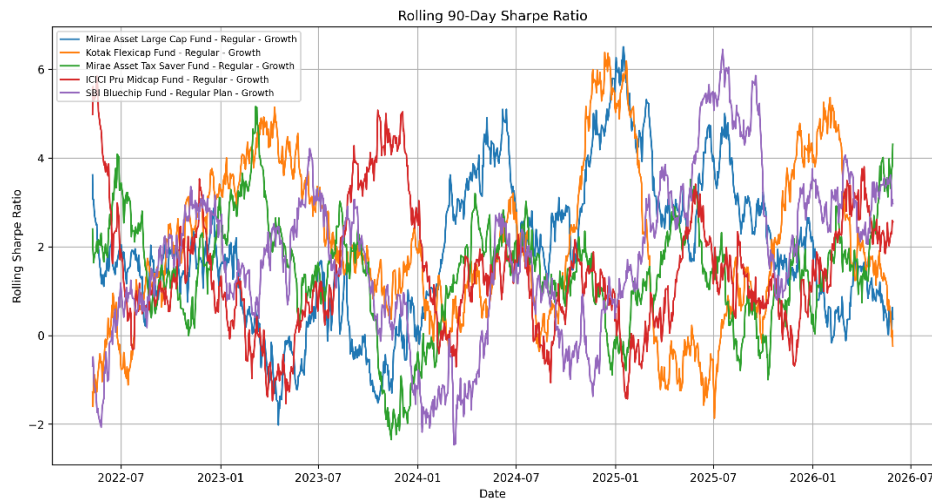


Figure 17: Rolling Sharpe Ratio of Selected Mutual Funds

Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Value at Risk (VaR) estimates the maximum expected portfolio loss over a specified confidence level during normal market conditions.

Conditional Value at Risk (CVaR), also known as Expected Shortfall, measures the average loss beyond the VaR threshold, providing a more comprehensive assessment of downside risk.

Both metrics help investors understand potential losses under adverse market conditions.

The calculated VaR and CVaR values were exported to a report for further analysis.

EXPLORER	ce_analytics.ipynb	README.md	var_cvar_report.csv	Advanced
MUTUAL_FUND_ANALYTICS				
data				
processed				
fund_performance_metrics.csv				
fund_scorecard.csv				
industry_folio_count_clean.csv				
investor_transactions_clean.csv				
monthly_sip_inflows_clean.csv				
nav_history_clean.csv				
portfolio_holdings_clean.csv				
scheme_performance_clean.csv				
sector_hhi_report.csv				
var_cvar_report.csv				
raw				
docs				
notebooks				
reports				
scripts				
sql				
queries.sql				
schema.sql				
data_dictionary.md				
README.md				
requirements.txt				

data	processed	var_cvar_report.csv
1	amfi_code,VaR_95,CVaR_95	
2	100016,-0.014363639988111142,-0.01806020763217176	
3	100025,-0.003793246199522682,-0.004993763020560281	
4	100033,-0.019033535861558825,-0.023455762770703794	
5	101206,-0.013281661986281489,-0.017439425687129195	
6	101207,-0.026021248221061066,-0.03245905898704435	
7	101208,-0.0002686575130400159,-0.0004220828655857685	
8	102885,-0.01261262882069356,-0.01549040525311211	
9	102886,-0.01922028344426503,-0.023250860269221213	
10	102887,-0.015231617279157872,-0.019411182679158694	
11	118632,-0.013954045870018117,-0.017618579320162156	
12	118633,-0.013442525143024242,-0.016630178254004827	
13	118634,-0.025438111458963106,-0.032304072697818535	
14	118635,-0.012553697284754662,-0.01617696049480912	
15	118636,-0.003804146068944347,-0.004918918420478294	
16	119092,-0.013749592076363814,-0.017327870956436054	
17	119093,-0.014225516634796053,-0.01748670974901306	
18	119094,-0.018480276243938307,-0.02426005947837057	
19	119095,-0.02618842440959035,-0.03166729360590828	
20	119120,-0.003938283121850805,-0.005013693153022244	
21	119551,-0.012846228528842571,-0.01639680693408763	
22	119552,-0.013500970479935082,-0.017335699664199813	
23	119598,-0.0245070519056771,-0.03059526410880242	
24	119599,-0.02685943514961382,-0.03238412245049217	
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26	120504,-0.013728193295939772,-0.017494143685606606	

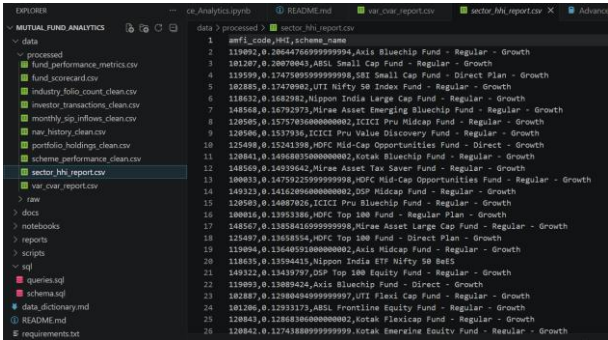
Table 11.1: Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Sector Concentration Analysis (HHI)

The Herfindahl–Hirschman Index (HHI) was used to measure the concentration of sector allocations within mutual fund portfolios.

Higher HHI values indicate greater concentration in a limited number of sectors, while lower values represent better diversification.

This analysis assists investors in understanding portfolio concentration risk and identifying highly diversified mutual fund schemes.



The screenshot shows a Jupyter Notebook with a file explorer on the left and a code cell on the right. The file explorer lists files like 'data', 'processed', 'fund_performance_metrics.csv', 'fund_scorecard.csv', 'industry_folio_count_clean.csv', 'investor_transactions_clean.csv', 'monthly_sip_inflow_clean.csv', 'nav_history_clean.csv', 'portfolio_holdings_clean.csv', 'scheme_performance_clean.csv', 'sector_hhi_report.csv', 'var_csr_report.csv', 'raw', 'notebooks', 'reports', 'scripts', 'queries.sql', 'schema.sql', 'data_dictionary.md', 'README.md', and 'requirements.txt'. The code cell displays a list of mutual funds with their HHI values and sector allocations.

Index	HHI	Fund Name	Sector Allocation
1	0.2864476699999999	Axis Bluechip Fund - Regular - Growth	Regular - Growth
2	0.28078043	ABSL Small Cap Fund - Regular - Growth	Regular - Growth
3	0.12678089999999999	SBI Small Cap Fund - Direct Plan - Growth	Direct Plan - Growth
4	0.127470982	UTI Nifty 50 Index Fund - Regular - Growth	Regular - Growth
5	0.1682982	Nippon India Large Cap Fund - Regular - Growth	Regular - Growth
6	0.16792973	Mirae Asset Emerging Bluechip Fund - Regular - Growth	Regular - Growth
7	0.15778300000000002	ICICI Pru Midcap Fund - Regular - Growth	Regular - Growth
8	0.1577936	ICICI Pru Value Discovery Fund - Regular - Growth	Regular - Growth
9	0.15241398	HFCL Mid-Cap Opportunities Fund - Direct - Growth	Direct - Growth
10	0.14960810000000002	Kotak Bluechip Fund - Regular - Growth	Regular - Growth
11	0.1493042	Mirae Asset Tax Saver Fund - Regular - Growth	Regular - Growth
12	0.14750225999999998	HFCL Mid-Cap Opportunities Fund - Regular - Growth	Regular - Growth
13	0.14120960000000002	DSP Midcap Fund - Regular - Growth	Regular - Growth
14	0.14007026	ICICI Pru Bluechip Fund - Regular - Growth	Regular - Growth
15	0.13953389	HFCL Top 100 Fund - Regular Plan - Growth	Regular Plan - Growth
16	0.13858416999999999	Mirae Asset Large Cap Fund - Regular - Growth	Regular - Growth
17	0.13858554	HFCL Top 100 Fund - Direct Plan - Growth	Direct Plan - Growth
18	0.13640591000000002	Axis Midcap Fund - Regular - Growth	Regular - Growth
19	0.13594418	Nippon India ETF Nifty 50 BeES	ETF Nifty 50 BeES
20	0.13439797	DSP Top 100 Equity Fund - Regular - Growth	Regular - Growth
21	0.13409425	Axis Bluechip Fund - Direct - Growth	Direct - Growth
22	0.13384049999999997	UTI Fund Cap Fund - Regular - Growth	Regular - Growth
23	0.12933173	ABSL Frontline Equity Fund - Regular - Growth	Regular - Growth
24	0.12868386000000002	Kotak Flexicap Fund - Regular - Growth	Regular - Growth
25	0.12743880999999999	Kotak Emerging Equity Fund - Regular - Growth	Regular - Growth

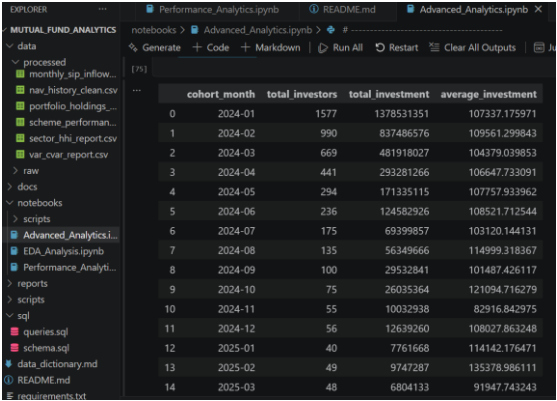
Table 11.2: Sector Concentration using Herfindahl–Hirschman Index (HHI)

Cohort Analysis

Investor Cohort Analysis was performed by grouping investors according to the month of their first investment.

This analysis provides insights into investor acquisition trends and helps evaluate how different investor groups behave over time.

Cohort analysis enables fund managers to identify seasonal investment patterns and monitor long-term investor engagement.



The screenshot shows a Jupyter Notebook with a file explorer on the left and a code cell on the right. The file explorer lists files like 'data', 'processed', 'monthly_sip_inflow...', 'nav_history_clean.csv', 'portfolio_holdings...', 'scheme_performanc...', 'sector_hhi_report.csv', 'var_csr_report.csv', 'raw', 'notebooks', 'scripts', 'Advanced_Analytics.i...', 'EDA_Analysis.ipynb', 'Performance_Analyti...', 'reports', 'scripts', 'queries.sql', 'schema.sql', 'data_dictionary.md', 'README.md', and 'requirements.txt'. The code cell displays a table with investor cohort data.

cohort_month	total investors	total investment	average investment
0	1577	1378531351	107337.175971
1	990	837486576	109561.299843
2	669	481918027	104379.039853
3	441	293281266	106647.733091
4	294	171335115	107757.933962
5	236	124582926	108521.712544
6	175	69399857	103120.144131
7	135	56349666	114999.318367
8	100	29532841	101487.426117
9	75	26035364	121094.716279
10	55	10032938	82916.842975
11	56	12639260	108027.863248
12	40	7761668	114142.176471
13	49	9747287	135378.986111
14	48	6804133	91947.743243

Table 11.3: Monthly Investor Cohort Analysis

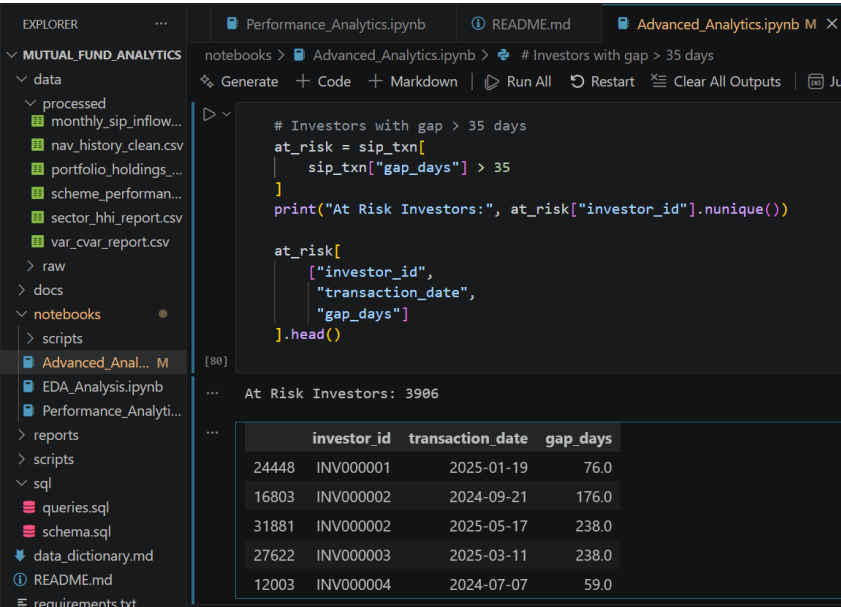
Investor Churn Analysis

Investor transaction history was analyzed to identify investors who had not made transactions for extended periods.

An investor was classified as "At Risk" if the inactivity period exceeded the defined threshold.

This analysis enables early identification of potential customer churn and supports targeted investor retention strategies.

The analysis identified **3,906 At-Risk Investors**, highlighting opportunities for proactive engagement.



```
# Investors with gap > 35 days
at_risk = sip_txn[
    sip_txn["gap_days"] > 35
]
print("At Risk Investors:", at_risk["investor_id"].nunique())

at_risk[
    ["investor_id",
     "transaction_date",
     "gap_days"]
].head()
```

At Risk Investors: 3906

	investor_id	transaction_date	gap_days
24448	INV000001	2025-01-19	76.0
16803	INV000002	2024-09-21	176.0
31881	INV000002	2025-05-17	238.0
27622	INV000003	2025-03-11	238.0
12003	INV000004	2024-07-07	59.0

Table 11.4: Sample of At-Risk Investors

Mutual Fund Recommendation System

A rule-based recommendation engine was developed to suggest mutual fund schemes based on their historical risk-adjusted performance.

The recommendation considers:

- Risk Grade
- Sharpe Ratio
- Annual Return

- Annual Volatility

Funds demonstrating strong historical returns with relatively lower risk are prioritized.

This module illustrates the practical application of financial analytics in assisting investment decision-making.

```

EXPLORER    ...    Performance_Analytics.ipynb    README.md    Advanced_Analytics.ipynb M
MUTUAL_FUND_ANALYTICS
├── reports\charts
│   ├── sortino_ratio.png
│   ├── state_sip.png
│   ├── top5_funds_compari...
│   └── top5_funds_compari...
├── scripts
│   ├── clean_aum.py
│   ├── clean_benchmark.py
│   ├── clean_category_inflo...
│   ├── clean_fund_master.py
│   ├── clean_industry_folio.py
│   ├── clean_nav_history.py
│   ├── clean_performance.py
│   ├── clean_portfolio.py
│   ├── clean_sip_inflows.py
│   ├── clean_transactions.py
│   ├── create_database.py
│   ├── data_ingestion.py
│   ├── data_quality_report.py
│   ├── fund_master_analysis...
│   ├── live_nav_fetch.py
│   └── recommender.py M
├── sql
├── OUTLINE
└── TIMELINE

scripts > recommender.py > ...
16  def assign_risk(volatility):
22      return "High"
23
24  scorecard["risk_grade"] = scorecard["annual_volatility"].apply(as
25
26  # -----
27  # User Input
28  # -----
29  risk = input("Enter Risk Appetite (Low / Moderate / High): ").str
30
31  # -----
32  # Filter Funds
33  # -----
34  recommended = (
35      scorecard[scorecard["risk_grade"] == risk]

PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    JUPYTER

FileNotFoundError: [Errno 2] No such file or directory: 'D:\\Mutual_Fund_An
scripts/recommender.pyMutual_Fund_Analytics>
Enter Risk Appetite (Low / Moderate / High): Low

Top 3 Recommended Funds

                                scheme_name    ...    annual_volatility
0    Mirae Asset Large Cap Fund - Regular - Growth    ...    0.141937
4        SBI Bluechip Fund - Regular Plan - Growth    ...    0.137414
9    Nippon India Large Cap Fund - Regular - Growth    ...    0.141484

[3 rows x 5 columns]
PS D:\Mutual_Fund_Analytics>

```

Table 11.5: Recommended Mutual Funds Based on Risk Profile

11. Power BI Dashboard

Dashboard Overview

To provide an interactive and user-friendly visualization of the mutual fund analytics, a Power BI dashboard was developed using the processed datasets generated during the Python analytics phase. The dashboard consolidates industry statistics, investor behavior, fund performance, portfolio insights, and SIP trends into a unified reporting interface.

The dashboard consists of four interactive pages that allow users to filter, compare, and analyze mutual fund data using slicers, KPI cards, charts, and tables. This enables fund managers, analysts, and investors to make informed decisions through dynamic visualizations.

Dashboard Features

The dashboard includes the following interactive features:

- KPI cards for summarizing important business metrics.
- Interactive slicers for filtering fund houses, categories, plans, states, and investor demographics.
- Line charts for trend analysis.
- Bar charts for comparative analysis.
- Scatter plots for risk-return visualization.
- Donut charts for distribution analysis.
- Matrix visualizations for monthly SIP comparisons.
- Dynamic tables displaying detailed fund and investor information.

Industry Overview Dashboard

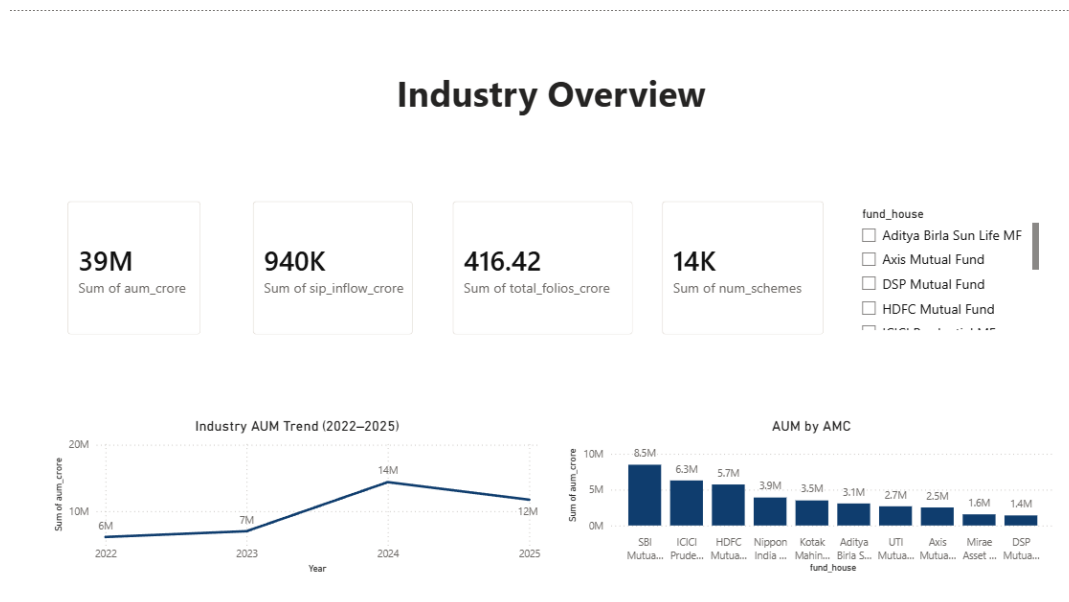


Figure 18: Industry Overview Dashboard

The Industry Overview dashboard presents a high-level summary of the Indian mutual fund industry by displaying key performance indicators and fund house statistics.

The dashboard highlights:

- Total Assets Under Management (AUM)
- Total SIP Inflows
- Total Folios
- Number of Mutual Fund Schemes
- Annual AUM Growth Trend (2022–2025)
- Assets Under Management by Asset Management Company (AMC)

Interactive filters allow users to analyze industry statistics for individual fund houses.

Key Insights

- The mutual fund industry experienced significant growth in AUM during 2024 before stabilizing in 2025.
- SBI Mutual Fund manages the highest AUM among the selected AMCs.
- The dashboard provides an overall snapshot of industry size and growth trends.

Fund Performance Dashboard

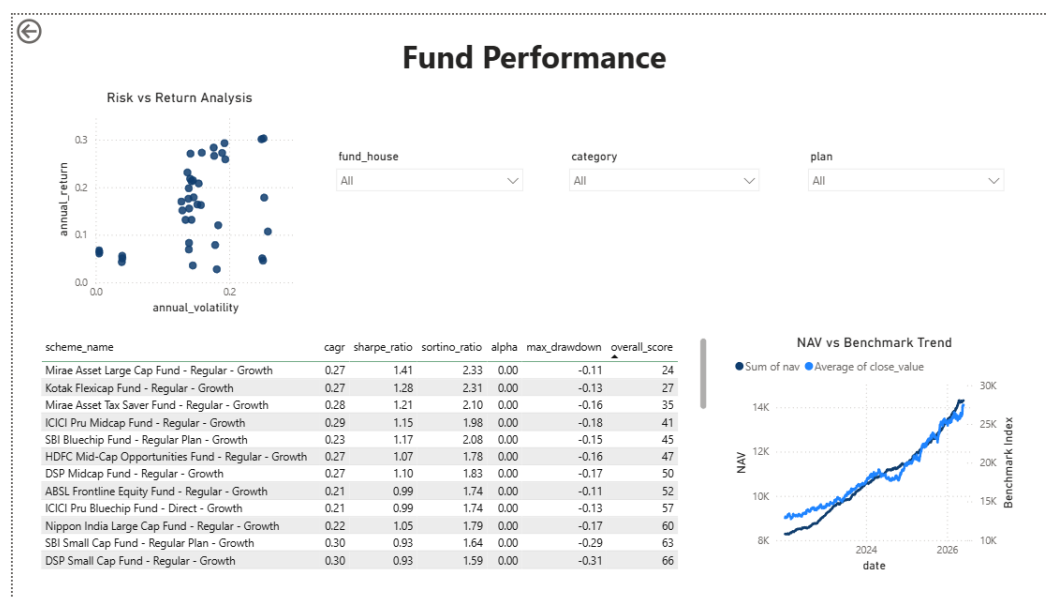


Figure 19: Fund Performance Dashboard

This dashboard evaluates mutual fund performance using multiple risk-adjusted return metrics. It enables users to compare schemes based on historical returns, volatility, and benchmark performance.

The dashboard includes:

- Risk versus Return scatter plot
- Fund performance score table
- NAV versus Benchmark trend comparison

- Filters for Fund House, Category, and Investment Plan

Key Insights

- The scatter plot illustrates the relationship between annual return and annual volatility across different schemes.
- The performance score table ranks mutual funds using CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- Benchmark comparison helps assess fund performance relative to market indices.

Investor Analytics Dashboard

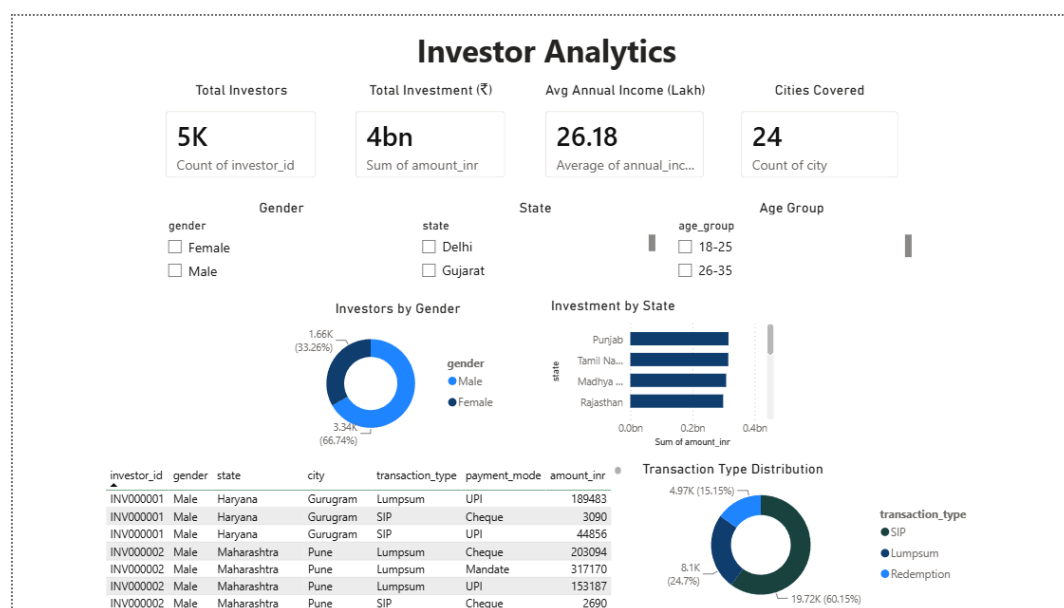


Figure 20: Investor Analytics Dashboard

The Investor Analytics dashboard focuses on investor demographics, investment behavior, and transaction patterns.

The dashboard provides information on:

- Total Investors
- Total Investment Amount

- Average Annual Income
- Cities Covered
- Gender Distribution
- State-wise Investments
- Transaction Type Distribution
- Investor Details Table

Interactive filters enable users to analyze investors by gender, age group, and state.

Key Insights

- Male investors constitute the majority of investors in the dataset.
- SIP transactions represent the largest proportion of investment transactions.
- Investment activity varies significantly across different states.

SIP & Market Trends Dashboard

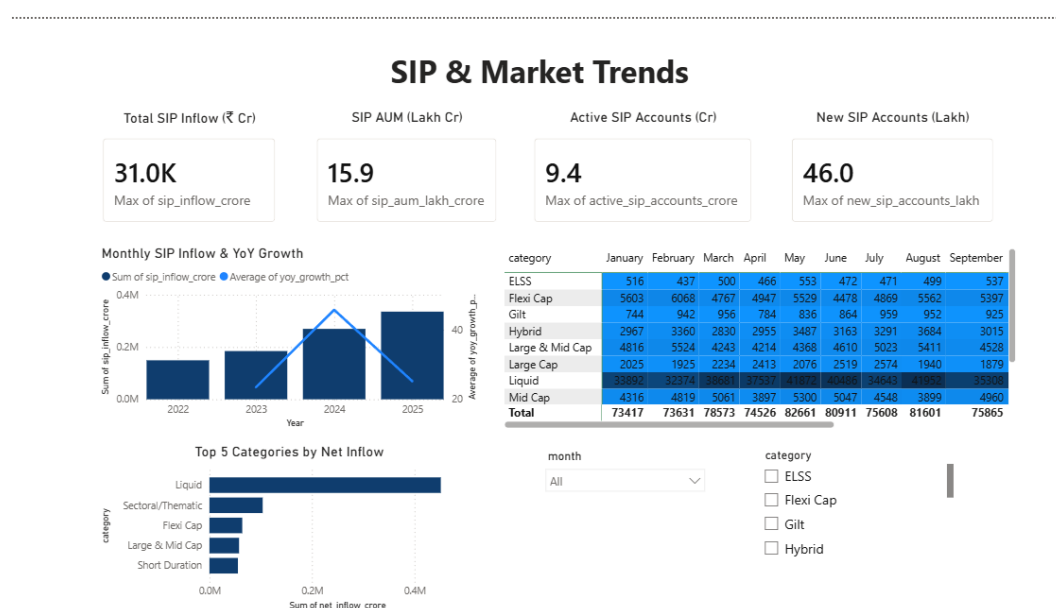


Figure 21: SIP & Market Trends Dashboard

This dashboard analyzes Systematic Investment Plan (SIP) growth and category-wise investment trends over time.

The dashboard contains:

- Total SIP Inflow
- SIP AUM
- Active SIP Accounts
- New SIP Accounts
- Monthly SIP Inflow with Year-over-Year Growth
- Monthly Category-wise SIP Matrix
- Top Categories by Net Inflow

Interactive filters allow users to explore monthly and category-wise SIP trends.

Key Insights

- SIP inflows have shown steady growth over the observed period.
- Liquid and Large Cap categories attract the highest net inflows.
- Monthly investment patterns indicate increasing participation through SIPs.

Overall Dashboard Insights

The Power BI dashboard provides a comprehensive visualization platform for mutual fund analytics by integrating industry statistics, fund performance, investor demographics, and SIP trends into a single interface.

The interactive dashboards enable users to:

- Monitor overall industry growth.
- Compare mutual fund performance using risk-adjusted metrics.
- Analyze investor demographics and transaction behavior.
- Evaluate SIP growth and market participation.
- Support data-driven investment decisions through interactive business intelligence reports.

13. Conclusion

This project successfully developed a comprehensive Mutual Fund Analytics platform by integrating data engineering, database management, exploratory data analysis, performance analytics, advanced analytics, and business intelligence reporting. The project transformed raw mutual fund datasets into meaningful insights through systematic data cleaning, SQL database creation, Python-based analytics, and interactive Power BI dashboards.

Various analytical techniques such as NAV trend analysis, AUM growth analysis, investor demographics, SIP trend analysis, risk-adjusted performance evaluation, investor churn analysis, cohort analysis, and mutual fund recommendation were implemented to support data-driven investment decisions.

The interactive dashboards and analytical reports enable stakeholders to monitor industry trends, compare fund performance, understand investor behavior, and evaluate market dynamics effectively. Overall, the project demonstrates the practical application of data analytics and visualization techniques in the mutual fund domain.

14. Future Scope

The current project can be further enhanced by incorporating additional analytical and machine learning capabilities. Some potential future improvements include:

- Integration of live mutual fund data using real-time APIs.
- Machine learning models for predicting NAV movements and fund performance.
- Personalized mutual fund recommendation using investor profiles.
- Portfolio optimization based on modern portfolio theory.
- Risk prediction using advanced statistical and AI techniques.
- Deployment of dashboards using Power BI Service for real-time reporting.
- Development of a web-based analytics platform using Streamlit or Flask.
- Automated report generation and scheduled dashboard refreshes.

15. References

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